FCPCA -

Fuzzy Clustering of Multivariate Time Series based on PCA

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INTRODUCTION

Why cluster multivariate time series (MTS)?

- Reveals hidden patterns in finance, bioscience and environmental science
- MTS like EEG have many interdependent channels and varying lengths
- MTS drift through mixed states (e.g., alert to drowsy), which hard clustering erases

Limitations of previous work

- Most fuzzy methods handle only univariate series
- Existing MTS methods are computationally heavy beyond ~20–30 channels, assume equal lengths, ignore lagged cross-correlations, and need a preset m and S

Applications and validation

- Main application: EEG driver drowsiness, using 3-second multi-channel segments to separate alert from drowsy states
- Reveals transitional states that hard clustering would miss
- Potential use in adaptive driver-monitoring systems
- Also validated on simulated data and benchmark datasets (Japanese Vowels, Basic Motions, and others)

ALGORITHM

Preliminaries:
1) Standardise the data
2) Initialize Membership matrix U randomly

Notations

- MTS:Multivariate Time Series
- N: number of MTS | p: channels (30)| T_i: length of series i (384) | S: clusters
- l: lag (l = 1,...,L, L = 2) | m: fuzziness (m > 1)
- X_i: $T_i \times p$ (ith Time Series)| $\hat{\Gamma}_i(l)$: $p \times p$ | $\mathbf{\Gamma}_i(l)$: $2p \times 2p$
- U (u_{is}): N × S, membership of series i in cluster s, rows sum to 1
- $\tilde{\Sigma}_s(l)$: $2p \times 2p$ | $C_s(l)$: set of component axes
- $Y_i^s(l)$: reconstructed data
- $E_{is}(l)$, E_i, E: scalars (reconstruction errors)

1) Weighted covariance matrix

For each cluster {1,2...,S} compute

$$\tilde{\mathbf{\Sigma}}_s(l) = \frac{\sum_{i=1}^{N} u_{is}^m \hat{\mathbf{\Gamma}}_i(l)}{\sum_{i=1}^{N} u_{is}^m}$$

$$\hat{\mathbf{\Gamma}}_i(l) = \begin{pmatrix} \hat{\Gamma}_i(0) & \hat{\Gamma}_i(l) \\ \hat{\Gamma}_i(l)^\top & \hat{\Gamma}_i(0) \end{pmatrix}$$

$$\hat{\Gamma}_i(l) = \frac{1}{T_i - l} \sum_{t=l+1}^{T_i} (\boldsymbol{X}_{i,t} - \overline{\boldsymbol{X}}_i)(\boldsymbol{X}_{i,t-l} - \overline{\boldsymbol{X}}_i)^\top$$

2) PCA components

Perform SVD → Take top eigenvectors(95% variance) → Project data(current,lagged) on axes

$$\hat{\mathbf{X}}_i(l) = (\mathbf{X}^*_{i,t-l}, \mathbf{X}^*_{i,t}), \qquad \boldsymbol{Y}_i^s(l) = \hat{\boldsymbol{X}}_i(l)\boldsymbol{C}_s(l)\boldsymbol{C}_s(l)^\top$$

3) Reconstruction error

For each cluster {1,2...,S} compute :

$$E_{is}(l) = \|\hat{\boldsymbol{X}}_i(l) - \boldsymbol{Y}_i^s(l)\|_2. \rightarrow E_i = \min_{s \in \{1,\ldots,S\}} \sum_{l=1}^{L} E_{is}(l), \rightarrow E = \sum_{i=1}^{N} E_i.$$

4) Update membership matrix:

$$u_{is} = \left[\sum_{s^*=1}^{S} \left(\frac{\sum_{l=1}^{L} \|\hat{\boldsymbol{X}}_i(l) - \hat{\boldsymbol{X}}_i(l)\mathbf{C}_s(l)\mathbf{C}_s(l)^\top\|^2}{\sum_{l=1}^{L} \|\hat{\boldsymbol{X}}_i(l) - \hat{\boldsymbol{X}}_i(l)\mathbf{C}_{s^*}(l)\mathbf{C}_{s^*}(l)^\top\|^2} \right)^{\frac{1}{m-1}} \right]^{-1}$$

Repeat until difference of error falls below tolerance

3 random restarts

METHODOLOGY

Choosing the right fuzziness index ***m*** and number of clusters ***S*** is crucial , an unsuitable pair can either under-partition the data or over-smooth the memberships. We use two CVIs to evaluate this.

Xie - Beni index

$$\mathrm{XB}_{S,m} = \frac{\sum_{i=1}^{N}\sum_{s=1}^{S} u_{is}^{m} \sum_{l=1}^{L} \left\| \hat{\boldsymbol{X}}_i(l) - \hat{\boldsymbol{X}}_i(l)\,\mathbf{C}_s(l)\,\mathbf{C}_s(l)^{\top} \right\|^2}{N \min\limits_{r \neq t} \sum_{l=1}^{L} \left\| P_r(l) - P_t(l) \right\|^2},$$

where

$$P_s(l) = \mathbf{C}_s(l)\,\mathbf{C}_s(l)^{\top}, \quad s \in \{1, \ldots, S\}.$$

Each $P_s(l)$ is a $2p \times 2p$ matrix. The set

$$\mathcal{P} = \left\{ P_s(l) \right\}, \quad s \in \{1, \ldots, S\},\ l \in \{1, \ldots, L\},$$

Fuzzy Silhouette index

$$z_i = \arg\min_{s} \sum_{l=1}^{L} E_{is}(l)$$

as the index of the cluster for which the projection of the ith time series gives the lowest total reconstruction error,

$$a_i = \sum_{l=1}^{L} E_{i\,z_i}(l)$$

as the intra-cluster reconstruction error of the ith time series, i.e., how well it fits its assigned cluster, and

$$b_i = \min_{s \neq z_i} \sum_{l=1}^{L} E_{is}(l),$$

the smallest reconstruction error for the ith time series among all other clusters excluding zi.

Let $ui(1)$ and $ui(2)$ denote the largest and second-largest membership degrees of the ith time series across all clusters.

$$\mathrm{FS}_{S,m} = \frac{\sum_{i=1}^{N} \left(u_{i(1)} - u_{i(2)} \right) \frac{b_i - a_i}{\max\{a_i, b_i\}}}{\sum_{i=1}^{N} \left(u_{i(1)} - u_{i(2)} \right)},$$

Grid search for optimal parameters

If the value of m approaches 1, the membership clusters approach binary set , and if it is high the clusters are uninformative. In order to select the value of m, we use grid search, selecting a value of m from the set [1.1 - 2.2] by running the algorithm across a grid and choosing the solution that minimizes XB and maximizes FS. When the two criteria disagree, we prioritize the lower XB, as it is directly aligned with the objective optimized by FCPCA.

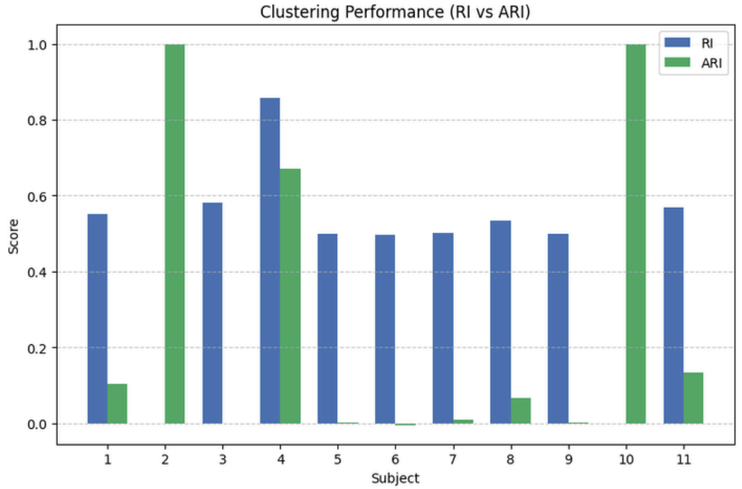
RESULTS

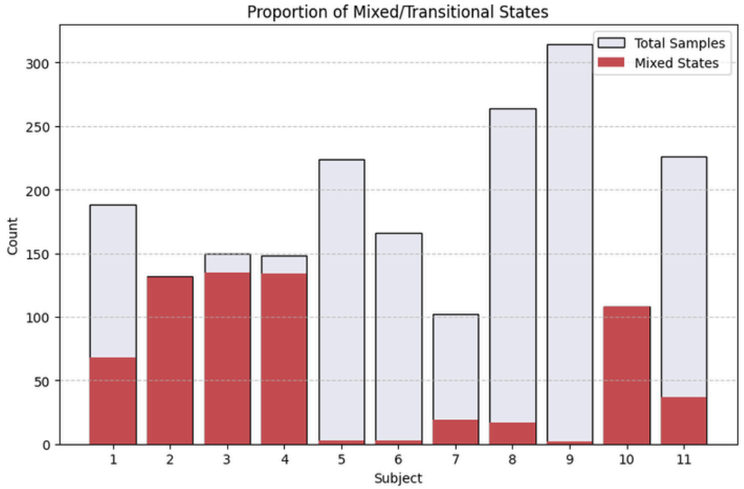
Evaluation criterion:

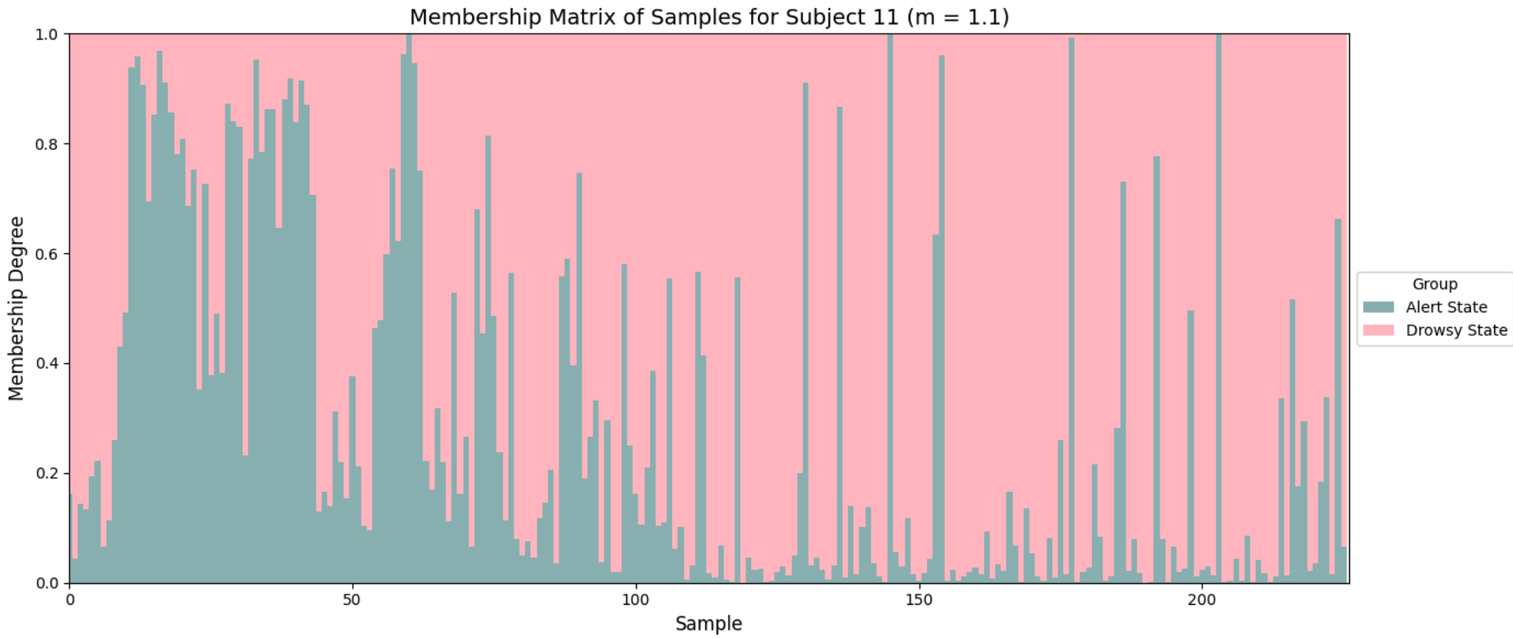
Rand Index (RI)

$$\mathrm{RI} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}},$$

where TP and TN are the numbers of correctly clustered and correctly separated pairs, respectively, while FP and FN represent misclustered pairs. RI has range [0, 1]. Higher RI values indicate greater similarity between the experimental and true partitions.







CONCLUSION

Results

- The FCPCA algorithm effectively handles multivariate time series by reducing dimensionality into common projection subspaces.
- When applied to the EEG dataset, the model successfully distinguished between alert and drowsy states while identifying mixed cognitive states.
- The methodology yielded strong clustering accuracy, measured via Rand Index (RI) and Adjusted Rand Index (ARI).

Limitations

- The method's reliance on estimating covariance matrices makes it sensitive to extreme noise, non-stationary segments, and outliers.
- The time lag structure must be predetermined manually, meaning a suboptimal choice could cause the model to miss important temporal dependencies.
- Computational complexity and memory footprint scale quadratically with the number of dimensions.

Future scope

- Developing robust extensions of FCPCA capable of automatically down-weighting to improve clustering reliability on contaminated data.
- Creating a mechanism to automatically select the optimal lag window from the data without manual tuning.
- Integrating the FCPCA framework into predictive modeling systems.
- Exploring approximate decomposition techniques to optimize computational efficiency

THANK YOU

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